MAPPA: Symmetry-Aware Atom Mapping through Continuous Fragment Growth

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Atom-to-atom mapping traces atom identities between molecular endpoints, supporting reaction analysis and pathway construction. Yet the same endpoints can admit multiple correspondences, and selecting a single answer can conceal alternatives with different chemical implications. We introduce MAPPA, a framework that efficiently discovers and retains alternative atom mappings. It uses shared molecular fragments and molecular symmetry to represent these alternatives compactly, allowing their bond changes to be examined and supporting subsequent chemistry-aware selection. MAPPA achieves the highest reference coverage among the evaluated methods on Golden. Standard benchmark runs recover 99.03% of Golden and 99.63% of EnzymeMap reference mappings, with matching and decoding together averaging 3.19 and 13.16 CPU seconds per reaction, respectively. Broader exploration raises Golden coverage to 99.51%. These results make broad exploration of mapping alternatives practical, providing a foundation for studying reaction ambiguity and incorporating chemical knowledge into mapping selection.

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1 Introduction

Atom-to-atom mapping (AAM) assigns a correspondence between atoms at the two molecular endpoints of a reaction. It describes the net evolution of atom identities through a chemical transformation, leaving the intervening elementary steps unspecified. Once this correspondence is established, changes in bonding can be identified explicitly (Figure 1). These assignments underpin reaction-center identification and the extraction of transformation templates used in reaction prediction and retrosynthesis (11; 1). They also connect endpoint geometries for reaction-path calculations (3). Reliable mapping therefore matters both for interpreting individual reactions and for building the datasets and models used to study chemical reactivity.

Endpoint structures, however, do not generally determine a unique correspondence. Molecular symmetry can make several atom labels structurally equivalent, while other assignments imply different atom origins and bond changes. Favoring fewer bond changes helps narrow the choices, but need not select the annotated mapping. In the EnzymeMap example in Figure 2, the reference places the formaldehyde carbon in the three-carbon molecule, whereas another correspondence assigns it to CO_2. Their heavy-atom counts differ, but both require six total bond events when hydrogen assignments are optimized separately for each correspondence. Both also account for the complete endpoint composition. Event counts therefore leave distinct atom origins unresolved in this example. Retaining alternatives allows subsequent interpretation to draw on additional chemical knowledge.

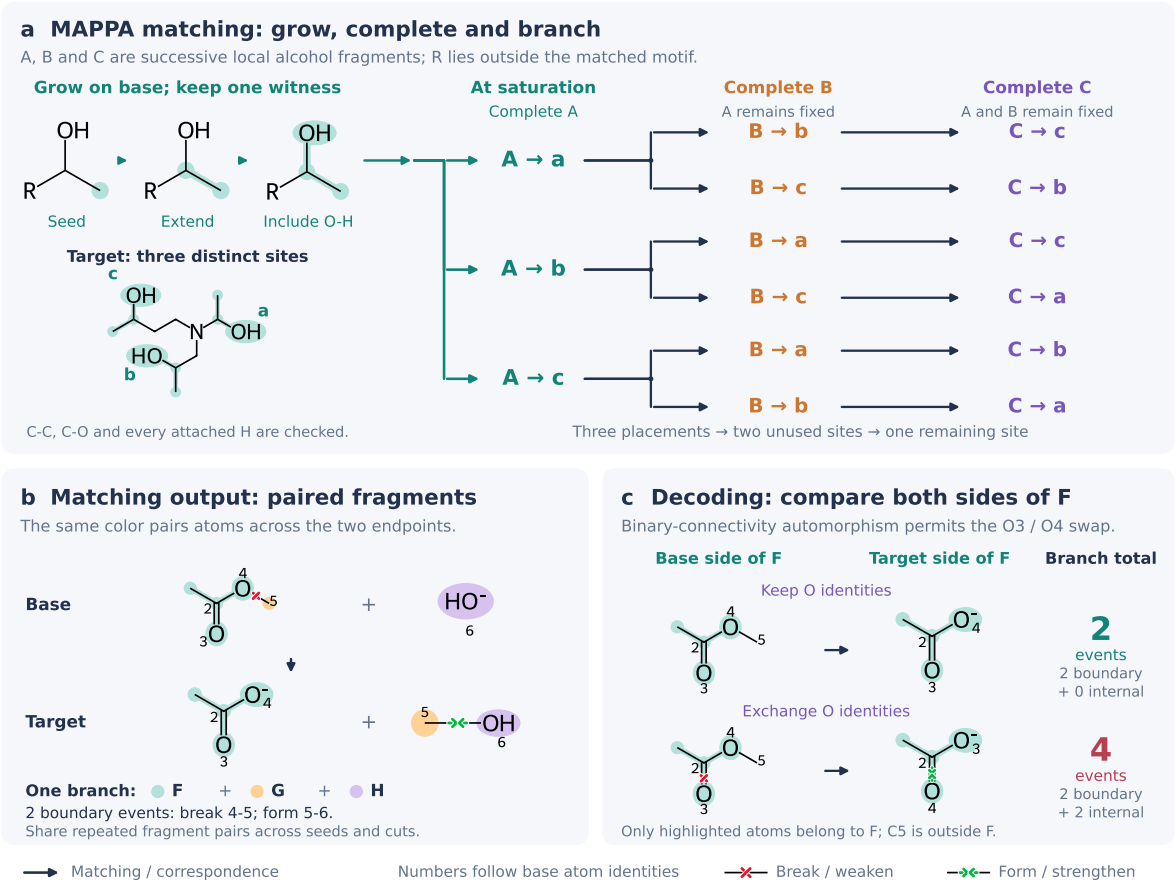


Figure 1 Fragment-based exploration of atom mappings with MAPPA. (a) Greedy growth and conditional placements. (b) Colors pair the base and target atom sets of fragments F, G and H. (c) An automorphism of F's binary connectivity permits O3 and O4 to exchange. Original bond orders then give two internal events, added to the two boundary events in b. Both endpoints are shown; unhighlighted C5 is outside F. Numbers follow base atom identities. Red marks indicate cleavage or weakening, green marks formation or strengthening. Constructed examples are detailed in Appendix A.

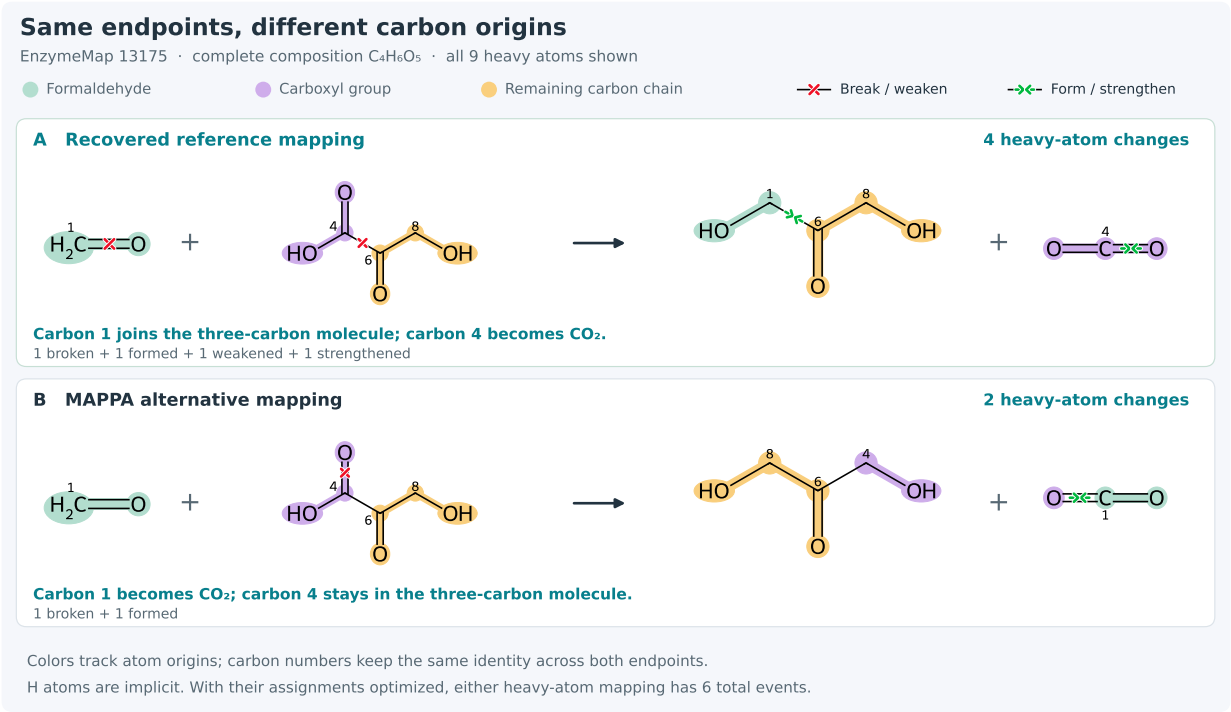


Figure 2 Distinct atom origins can share the same total event count. Two saved MAPPA heavy-atom correspondences for EnzymeMap case 13175 retain all nine heavy atoms and the complete composition $C_4H_6O_5$. A recovers the annotated mapping: formaldehyde carbon 1 enters the three-carbon molecule, while carboxyl carbon 4 becomes CO_2. B exchanges these carbon origins. The drawings show four and two heavy-atom bond changes, respectively; colors track atom origins and numbers identify the carbons. Minimizing hydrogen assignments separately for these fixed heavy-atom mappings adds two and four events, respectively, giving six total events in either case. Thus this example shows ambiguity under event counting, not a strict all-atom advantage over the reference. Hydrogen identities are not annotated by EnzymeMap. These correspondences do not establish reaction pathways. Provenance and counting details are given in Appendix A.4.

AAM methods can be organized by the output they seek to provide (Table 1). *Alternative-generating methods* retain several correspondences for subsequent interpretation or selection. SLAPMapper provides such alternatives through approximate graph matching, combining iterative refinement of atom labels with sequential linear assignment (7). *Prediction-oriented methods* aim to select a preferred correspondence. RXNMapper derives assignments from attention patterns in a reaction language model trained on unannotated reactions (11); LocalMapper learns from chemist-labeled examples through a human-in-the-loop procedure (1). These methods bring information from reaction data into the selection of atom correspondences.

Table 1 A taxonomy of AAM methods by output objective. Representative methods illustrate the distinction between retaining alternatives and selecting a preferred answer.

	Alternative-generating	**Prediction-oriented**
Objective	Preserve candidate correspondences for later interpretation or selection.	Identify the most plausible correspondence.
Output	Several mappings, stored individually or as compact families.	One preferred mapping.
Examples	SLAPMapper (7); MAPPA (this work).	RXNMapper (11); LocalMapper (1).
Evaluation	Reference coverage: is the annotated mapping retained?	Top-1 accuracy: does the selected mapping match the annotation?

The two objectives can be combined by placing a selector after an alternative-generating method. A selector might favor fewer bond events, apply chemical rules, or use a learned ranking. This separates the ability to discover a useful correspondence from the ability to choose it. Reference coverage measures the former; top-1 accuracy measures the latter. A single predicted mapping can also be evaluated as a candidate set of size one, providing a common coverage criterion across the categories.

For alternative generation, the computational challenge is to retain diverse correspondences without repeatedly searching and analyzing their shared structure. Choices among repeated groups can combine into many complete assignments, even when large parts of those assignments are identical. Enumerating each mapping separately repeats work and obscures which decisions distinguish the alternatives. An efficient representation must preserve those decisions while making their consequences for bond changes accessible. This motivates organizing the search around shared molecular fragments and their allowed rearrangements.

Here we introduce MAPPA, an alternative-generating mapper that treats atom correspondence as a constraint-search problem. Greedy growth discovers fragments compatible with both endpoints; placing one fragment restricts the choices available to the remaining atoms. Molecular symmetry represents rearrangements within these constraints compactly. A matching stage discovers fragment combinations, and a decoding stage relates their alternatives to bond events, reusing calculations across shared fragments. On Golden (8) and EnzymeMap (5), the standard benchmark runs achieve reference coverage of 99.03% and 99.63%, with matching and decoding together averaging 3.19 and 13.16 CPU seconds per reaction, respectively. Broader search raises Golden coverage to 99.51%. We also examine how many alternatives remain under a minimum-event preference, connecting coverage and computational cost to the subsequent task of choosing a mapping.

2 Methods

2.1 Overview: from shared fragments to alternative mappings

For each search direction, the *base* is the endpoint on which fragments grow and the *target* is where their placements are sought. Either endpoint can serve as the base; reversing the search exchanges these roles. MAPPA builds atom mappings by growing structural fragments shared between these endpoints (Figure 1). Starting from a seed atom, it extends a fragment while a compatible placement remains possible. When no admissible extension remains, it finds the completed fragment's alternative placements. Each placement defines a search branch: that choice is locked, and fragment growth continues on the remaining atoms under the accumulated assignments. Repeating this process produces alternative compositions of matched fragments. Different seeds, single-bond cuts and search directions expose additional compositions.

Two stages organize this procedure. The *matching stage* discovers fragments and their conditional placements, then reduces repeated fragments and branch compositions into a shared mapping catalogue. Reduction is its final subphase. The *decoding stage* uses that catalogue to relate permitted atom rearrangements to bond events, preparing each unique fragment once and combining its response with the branch boundaries. Its output retains branches that can be queried within a chosen event-count window, rather than only one selected mapping. Matching and decoding are separate operations; a saved catalogue can be decoded without repeating discovery. Reference verification evaluates these outputs separately (Section 3.2).

2.2 Matching Stage: grow, saturate and branch

Matching alternates between *growth matching*, which tests for a relaxed subgraph isomorphism, and *saturation matching*, which finds the completed fragment's distinct target atom-set placements. Postponing branching lets a small, common motif grow into a larger, more restrictive constraint (Figure 1a).

2.2.1 Growth matching: keep extending while a match exists

The growing fragment $g = (B_g, A_g)$ contains base atoms B_g and accepted edges A_g; $\mathcal{L}$ records earlier locked assignments. A *relaxed subgraph isomorphism* maps these atoms injectively into the target, preserving elements and locks, with each accepted edge placed on a target bond within tolerance τ_{iso}. A *witness* m is

one such map. Explicit hydrogens participate. Matching is non-induced: extra target edges are allowed, and unselected base bonds impose no additional constraint.

Growth starts from a seed with an available target atom. The presence threshold b defines eligible connections. It selects a boundary atom uniformly, then one of its frontier edges uniformly, and tentatively forms g^+. The decision is an existence test:

$$\text{Exists}(g^+ \mid \mathcal{L}) = \mathbf{1}\big[\mathcal{M}(g^+ \mid \mathcal{L}) \neq \varnothing\big] , \tag{1}$$

where $\mathcal{M}$ is the set of relaxed subgraph isomorphisms satisfying the locks. If a witness is found, accept the extension. If no placement exists, defer that edge, keep g, and try the remaining frontier. *Saturation occurs only when no eligible extension remains.* One failed edge does not end growth.

Let $\nu(g)$ count distinct target atom sets admitting a placement (Table 2). Growth does not compute this count: every positive outcome means continue growing, regardless of multiplicity.

Table 2 Match-count transitions during proposed growth. "Many" means more than one target atom-set placement, not multiple symmetry permutations on the same set. The $0 \to 0$ row is a rejected start; a zero-match fragment is never expanded.

$\nu(g) \to \nu(g^+)$	Decision
$0 \to 0$	No seed placement: skip this start and try another seed.
$1 \to 0$	Keep g and defer the edge. At saturation, one placement gives one continuation.
many $\to 0$	Keep g and defer the edge. At saturation, resolve its alternative placements.
$1 \to 1$	Accept the extension and continue growing.
$1 \to$ many	Accept; postpone the placement choice until saturation.
many $\to 1$	Accept; the larger fragment has narrowed the choices.
many $\to$ many	Accept; retain one witness and continue growing.

For $1 \to$ many, consider one eligible target carbon with two oxygen neighbors. C has one target atom set; C–O has two. Finding either placement accepts the extension without branching. The witness can change target sites during later growth, while earlier locks remain fixed. Extending the current witness or reusing a compatible saved fragment can avoid a new search; otherwise the existence solver finds one witness or establishes absence. A resource limit is not a zero-match answer.

2.2.2 Saturation matching: recover distinct atom-set placements

At saturation, MAPPA finds distinct target atom sets $T_i = m(B_g)$ reached by relaxed subgraph isomorphisms. Each pair (B_g, T_i) defines a completed fragment placement, with its internal mapping relation and the binary-graph automorphisms of its two sides retained compactly.

A six-carbon ring fragment fitting two target rings with different external attachments gives two atom-set placements (Figure 3). Ring rotations and reflections are fragment automorphisms: they change the correspondence within a placement without changing its atom sets. Search excludes already represented target sets directly, avoiding enumeration of their bijections followed by deduplication.

Fragment automorphisms act within each endpoint's fragment graph. *Global automorphisms* act on the whole target graph. A certified subgroup of the latter can share work between placements only if it preserves the entire remaining matching problem, moving later fragments jointly and respecting earlier locks. Otherwise those placements remain separate (Appendix A.1).

2.2.3 Hierarchical branching: lock a placement and continue

Each completed placement creates a child continuation: lock its assignments and grow the next fragment. In Figure 1a, choosing A$\to a$ leaves sites b, c for B; choosing B$\to b$ leaves c for C. These are conditional decisions: two fragments cannot independently occupy the same target atoms. Joining an earlier fragment retains its locked assignments and accepted constraints.

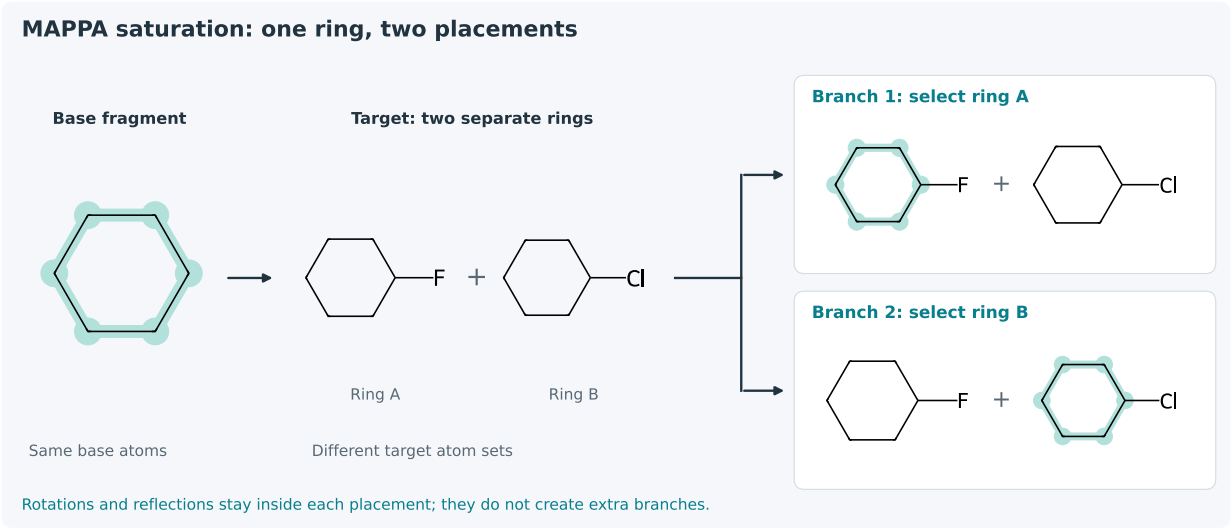


Figure 3 Two ring placements give two branches. A fixed six-carbon fragment fits either target ring. The different F and Cl attachments prevent a global automorphism from exchanging the rings. Each child selects one target carbon atom set, highlighted in teal; rotations and reflections within that ring remain inside the placement rather than creating additional branches. This is a carbon-fragment illustration with hydrogen constraints omitted, not a complete reaction.

Each locked fragment expresses a shared structural constraint, extended as far as the current greedy trajectory permits. Another seed may discover a different constraint. After each completed step, apply the branch cap from leaves to root: discard the largest sibling subtrees as needed, recount, then assess the parent. Internal symmetry permutations do not count as branches. Caps restrict coverage and are recorded explicitly (Appendix A.2).

2.3 Matching Stage: seeds, cuts and search direction

Vary the starting point. One greedy trajectory can repeatedly favor a large preserved region and miss another useful decomposition. In Figure 4a, the highlighted benzyl region offers more starting atoms than the remaining methylamine region. Additional seeds can discover different constraints, but may also revisit the same large fragment.

Vary the allowed connections. A single-edge *cut sweep* perturbs this commitment systematically. MAPPA searches the uncut graph and then repeats the search with each eligible base edge temporarily masked, including bonds to hydrogen. Each cut condition receives the requested number of seed orderings. Masking a connection inside the dominant region can expose a trajectory that ordinary seeding rarely reaches. The original endpoint weights remain unchanged for decoding: a mask alters discovery and does not assert a broken bond. Its endpoints may reconnect through another route, and one cut can redirect several later fragment choices.

Exchange the endpoints. Exchanging the base and target adds further alternatives. Seeds, cuts and reverse searches broaden discovery without making it exhaustive. The same exploration principle also broadens SLAP's returned alternatives: the expanded protocol described in Section 4.2 reaches approximately 97% coverage.

2.4 Matching Stage: consolidate unique fragments and branches

Different seeds and cuts often rediscover the same fragment. The final matching subphase, *reduction*, gives each base/target atom-set pair (B_i, T_i) one shared identity F_i. For example, two searches that match $\{1, 2, 3\}$ to $\{a, b, c\}$ reference the same F_i, even if their growth order or starting witness differs. Each branch then references its fragment identities, allowing repeated compositions to share one catalogue entry.

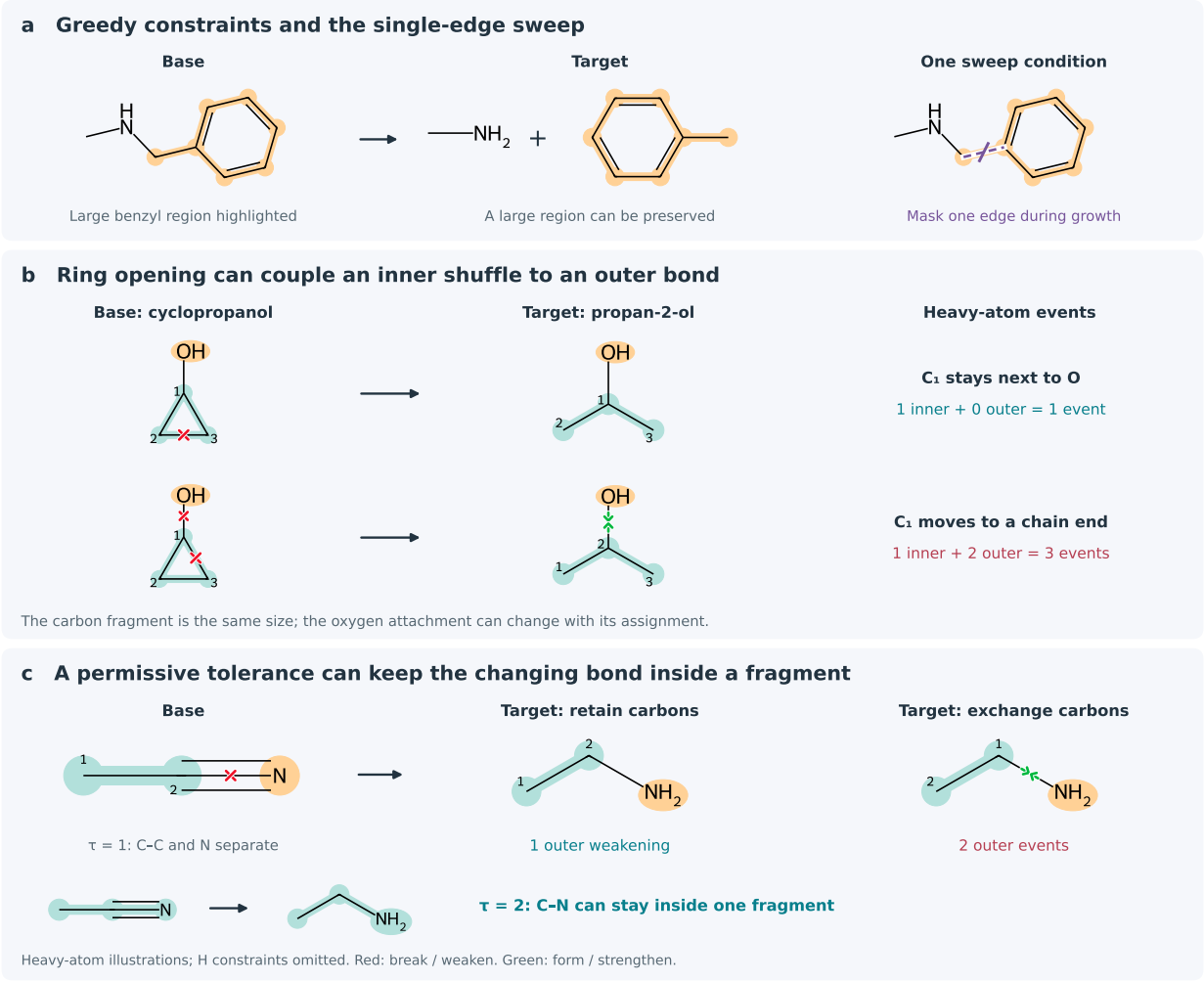


Figure 4 Molecular examples of exploration and boundary dependence. (a) The benzyl region is preserved between N-methylbenzylamine and the displayed methylamine/toluene endpoint. Masking one of its edges changes the growth conditions; the purple mark is a search mask, not a claimed cleavage. (b) In cyclopropanol and propan-2-ol, the teal carbon fragment has one internal ring-opening event under either displayed assignment. Carbon 1 at the chain center preserves its oxygen attachment; at an end, it adds an outer cleavage and formation. The heavy-atom totals are one and three. (c) For acetonitrile and ethylamine, tolerance one can leave C–C and N separate: exchanging the carbons changes one outer weakening into two outer events. Tolerance two can include the changing C–N bond inside one fragment. Numbers track the same carbon atoms; red and green marks indicate losses and gains as in Figure 1. These are heavy-atom graph illustrations, with hydrogen constraints omitted and event threshold 0.5; panels b and c are checked by direct pair counting. They illustrate possible boundary dependence, not its measured frequency.

A fragment identity is a key, not permission to map its atoms arbitrarily. Reduction keeps one original record with its witness, discovery constraints, anchors and symmetry relation. Branches also retain the ordered placement context needed to move later fragments consistently. Equal compositions are consolidated with that context preserved; their cut provenance is collected together. Selecting one original record does not retain the union of all discarded discovery relations.

This subphase reuses the fragment inventories already accumulated during matching. It performs structural grouping, with no symmetry optimization or event computation. Its output is one shared catalogue per reaction and direction. That catalogue is the input to decoding.

2.5 Decoding Stage: prepare each fragment's symmetry–event response

Decoding turns the retained mapping alternatives into queryable bond-event information. It first prepares each unique fragment once, then reuses that response wherever the fragment occurs. The local calculation connects *which atoms can move together* to *which bond changes those moves produce*.

2.5.1 Fragment automorphisms: independent and correlated moves

A *fragment automorphism* permutes atoms within one endpoint's fragment while preserving its binary connectivity and element labels, with anchors fixed. We compute these automorphisms separately for the base and target sides. Bonds satisfying the presence criterion share one structural label; original bond weights are retained for event counting. External attachments do not constrain the fragment automorphisms.

Nauty supplies compact generators of these automorphism groups (9): unit operations that describe the allowed rearrangements. Exchanging two identical arms moves each arm with its attached atoms; exchanging two hydrogens can act within an arm. The two arms must still occupy different positions. These joint operations retain that correlation without storing every full permutation. Both endpoint actions are represented; for an aligned equal-connectivity pair, one side suffices to describe their relative rearrangements.

2.5.2 Internal events: attach bond changes to the allowed moves

Let W_B and W_T denote the original base and target bond-weight matrices. For mapped atoms u, v, an assignment m changes the bond weight by

$$\Delta W_{uv}(m) = W_T(m(u), m(v)) - W_B(u, v). \tag{2}$$

The pair contributes one event when $|\Delta W_{uv}|$ exceeds its event threshold. We use empirical bond-order-change thresholds of 0.5 for organic bonds (the default for non-metal pairs) and 0.3 for bonds involving at least one metal atom. The sign distinguishes formation or strengthening from cleavage or weakening. These thresholds differ from the growth tolerance: a binary symmetry can preserve connectivity while changing bond weights.

In the acetate example (Figure 1c), O3 and O4 are interchangeable under an automorphism of the fragment's binary connectivity: each oxygen is adjacent to the same carbon. Their original C–O bond orders differ, so exchanging their assignments adds one weakening and one strengthening, giving two internal events. The saved response associates the rearrangement with those changes, so every branch using that fragment can reuse the calculation. Independent symmetry components are processed separately when no event term or requested endpoint couples them. Event-invariant choices share computation; correlated choices remain joint. The response supports later event queries without expanding every complete mapping.

2.5.3 Different connectivity: improve the starting correspondence

The accepted growth graph can omit bonds present in the final fragment. In the substituted triangle–chain example (Figure 4b), the displayed local shuffles keep one internal ring-opening event while changing the outer attachment. More generally, two valid relaxed subgraph isomorphisms on the same atom sets need not be connected by the final fragment automorphisms. Optimizing only one starting correspondence can therefore miss a lower-event alternative.

MAPPA first improves the saved witness under fragment automorphisms. Additional starting-correspondence search is allowed only when growth and final connectivity differ on at least one side and the witness has not

attained an established event lower bound. Preparation and improvement share a fixed per-fragment budget. The output retains the selected starting correspondence and its symmetry response. An unsettled minimum is recorded as best found; selecting this starting orbit restricts decoded coverage. Solver details and bounds are given in Appendix A.3.

2.6 Decoding Stage: assemble branch events and select the window

A branch combines three contributions: internal fragment events, ordinary boundary events and the correction for the swept bond. For a branch $\mathcal{B}$ and a permitted correspondence m, the reported count is

$$E_{\mathcal{B}}(m) = B_{\mathcal{B}} + \sum_i e_i(m) + s_{\mathcal{B}}(m), \qquad 0 \le s_{\mathcal{B}}(m) \le 2. \tag{3}$$

Here e_i is the reused internal response of a disjoint final fragment piece, $B_{\mathcal{B}}$ is the ordinary-boundary contribution, and $s_{\mathcal{B}}$ accounts for the swept connection. The following rules specify how they are combined.

2.6.1 Ordinary boundaries: a fixed contribution under the model

MAPPA uses the independent ordinary-boundary approximation: these contributions remain fixed as internal fragment choices change. Bonds touching mapped atoms are counted, including mapped/unmapped boundaries; bonds wholly within unmatched material are omitted. No assignment is invented for an unmatched atom.

Greedy growth helps absorb connections that would otherwise couple fragments. In C–C≡N → C–C–N (Figure 4c), tolerance one can leave C–C separate from N. Swapping the carbons changes the outer contribution from one weakening to a cleavage plus a formation. Tolerance two allows C–N inside the fragment, where the internal response handles that dependence. Greediness helps, but does not guarantee independence: the triangle–chain example still has a variable outer attachment. Equation (3) is therefore a model count, which can differ from the physical whole-reaction count.

Random growth, multiple seeds and the cut sweep can mitigate this limitation by discovering alternative fragment decompositions. In Figure 4b, growing through the O–C attachment first can place it inside a shared fragment instead of leaving it across a boundary. Another branch can then represent the same correspondence with its correct event count, even if the original branch assigns it differently. This exploration does not repair the original branch's approximation or guarantee a correct representation elsewhere. The benchmarks do not isolate the size of this mitigation.

2.6.2 The swept bond: couple neighboring fragment responses

A sweep masks one base bond. When its endpoints lie in different final fragments, those fragments are neighbors across the masked connection. Their internal event responses can be prepared separately, but this boundary contribution depends jointly on their choices: preserving the connection requires the two atoms to occupy compatible target positions. The swept bond therefore introduces a coupling between neighboring fragments that the decoder handles explicitly.

During branch assembly, MAPPA reuses the prepared responses to evaluate the two endpoint positions together, retaining their dependence on internal events and one-to-one occupancy. If both endpoints remain in one fragment, the same query uses that fragment's response. The associated contribution $s_{\mathcal{B}}$ ranges from zero to two events: a correspondence can preserve the connection, produce one bond change, or break one connection and form another. The sweep still masks only one bond; the event count describes the resulting correspondence across the two molecular endpoints.

2.6.3 The event window: retain mappings near the minimum

The event window focuses decoding on mappings with few bond changes. MAPPA first adds the local fragment minima and fixed boundary contribution for each branch. Let L be the smallest of these totals before resolving the swept bond. When the fragment minima can be combined, the sweep adds at most two events, giving an initial branch count no greater than $L + 2$. Only branches whose lower bounds are at

most $L + 2$ need to be examined for a lower count. Their swept-bond contributions are evaluated using the prepared fragment responses.

The lowest count found, U, sets the requested window $[U, U + n]$. MAPPA retains branches that could contribute mappings within this window and excludes any branch whose lower bound exceeds $U + n$. The studies here use $n = 5$. Retained branches reference the shared fragment responses, allowing subsequent queries for mappings at a chosen event count or with specified atom correspondences.

2.7 Execution: parallel stages and shared computation

The stage boundaries also define parallel work. Reactions and directions are independent; seed/cut searches within a direction can run concurrently before their results are reduced together. Growth along one branch remains sequential because later fragments depend on earlier occupancy. For the reported batch workflow, the matching stage, including reduction, finishes before catalogue decoding begins. Different reaction/direction catalogues can then be decoded independently, with unique-fragment responses shared within each catalogue.

A persistent worker pool schedules these operations through the same interfaces used by direct local calls. Intermediate records can remain in memory or use aggregated storage. Compatible fragment proofs can be reused during growth, but each seed and cut keeps its own random decisions. Matching and decoding use a shared symmetry core, and bulk operations use native kernels. Scheduling and persistence do not determine which mappings are allowed. Algorithm details and pseudocode appear in Appendix A; execution and storage interfaces are documented in the repository.

3 Evaluation

3.1 Datasets and search settings

We evaluate all 1,851 Golden reactions (8) and the prepared collection of 47,974 EnzymeMap reactions (5). Reference labels are removed before search. Both studies use the prepared formal-bond-order matrices and explicit hydrogens. EnzymeMap supplies curated database mappings as references.

The common settings are discovery tolerance $\tau_{\text{iso}} = 2$, presence/cut threshold $b = 0.2$, hierarchical branch cap 100, random seed 42, both directions and the full single-edge sweep. Golden uses one and ten seed orderings per cut; EnzymeMap uses one. Decoding uses the original bond orders and empirical event thresholds of 0.5 for organic bonds and 0.3 for bonds involving a metal, with a five-second allowance per unique fragment and the best attained model count plus five. No reference mapping guides growth, pruning or starting-correspondence selection.

We compare one-seed MAPPA with two SLAPMapper configurations on identical Golden inputs and the same CPU model. Standard SLAP uses the uncut binary graph; expanded SLAP combines binary and weighted modes with the full single-edge cut sweep. All three configurations run in both directions; the paired study supplies the computational-cost comparison.

3.2 Reference verification

Coverage measures whether the annotated correspondence is present in MAPPA's decoded output within the minimum-plus-five event window. A recovered reference requires a concrete one-to-one mapping from a retained branch and its allowed symmetry choices. We replay that mapping and independently check its event count.

For both datasets, the check preserves annotated heavy-atom identities and unmatched-heavy-atom requirements; unannotated hydrogens must admit a feasible completion. Recovery in either search direction counts once. This measures reference inclusion among alternatives, not top-ranked prediction accuracy. Every reaction remains in the denominator. Comparator evaluations additionally allow chemically equivalent endpoint labels, preserving charge, isotope, hydrogen count and stereochemistry.

3.3 Timing and reference-event analysis

Matching time includes both fragment discovery and reduction into the final catalogue; decoding is reported separately. CPU seconds sum successful worker computation over both directions and all cuts, including stage input loading and result construction. Publication and external verification are excluded. Source identities, hardware and completed-pass accounting are documented in the supplement.

To locate the reference relative to the detected minimum, let m_{GT} be a saved, verified mapping that agrees with the annotated atom pairs. We measure

$$n_{\mathrm{GT}} = E(m_{\mathrm{GT}}) - U, \tag{4}$$

where U is the detected minimum already saved by decoding, taking the lower count across both directions. When both directions supply verified mappings, we use the one with the lower recorded count. Both terms use the same all-atom decoder model. Unannotated atoms, including hydrogens, retain their assignments in the verified witness. This comparison uses existing counts without further optimization.

For references absent from the saved output, we fix their annotated pairs as anchors and complete the unspecified assignments with one seed per direction and no cut sweep. We compare the lowest attained completion count across both directions with the original U. The supplied pairs remain fixed, and the completed mapping is checked against the original bond matrices. These completion offsets are added to the same distribution, so that every reaction contributes once. Anchored completions do not contribute to independent mapping coverage.

We separately count the distinct reaction cores observed among saved mappings at the detected minimum. A core records which bonds decrease or increase in order, including bonds to hydrogen. Symmetry-equivalent atom labels count once, and different fragment partitions with the same bond-change pattern share one core. We combine both search directions in a common atom-index frame and evaluate changes directly from the original endpoint matrices. This measures the chemical differences between recorded alternatives without searching for additional symmetry arrangements.

4 Results

4.1 Reference coverage across datasets and methods

MAPPA recovers 1,833 of 1,851 Golden references with one seed (99.028%) and 1,842 with ten seeds (99.514%). On EnzymeMap, one seed recovers 47,796 of 47,974 references (99.629%; Table 3 and figure 5). These references occur within the decoded minimum-plus-five event window. Matching, decoding and reference evaluation are complete for both datasets.

Table 3 Reference coverage on Golden and EnzymeMap. A reference is recovered when a verified mapping occurs within the decoded minimum-plus-five event window in either direction. Percentages use all reactions.

Study	Reactions	Recovered	Not recovered
Golden, 1 seed	1,851	1,833 (99.028%)	18
Golden, 10 seeds	1,851	1,842 (99.514%)	9
EnzymeMap, 1 seed	47,974	47,796 (99.629%)	178

Ten seeds recover nine additional Golden references without losing any recovered by one seed. Wider exploration therefore improves coverage while retaining the same decoding window.

Table 4 compares Golden coverage across methods. A single prediction is treated as a candidate set of size one. MAPPA achieves the highest coverage among the evaluated methods. LocalMapper has the highest coverage among the evaluated single-answer interfaces at 86.71%, followed by Chython at 84.33% and RXNMapper at 83.58% (1; 2; 11). Default binary SLAP covers 85.95%; combining binary and weighted modes with both

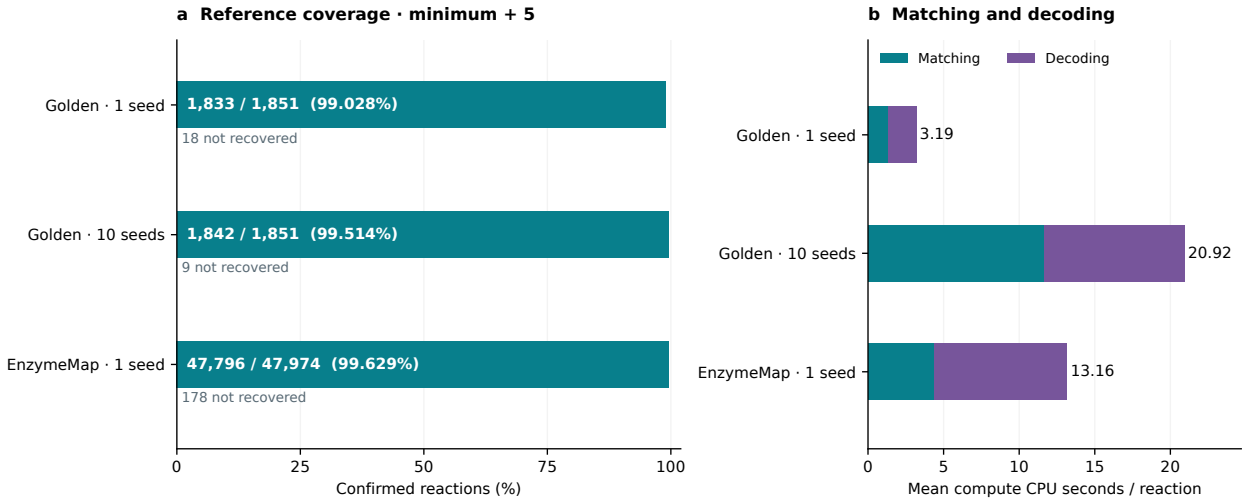


Figure 5 Reference coverage and computational cost. (a) Reference inclusion in the decoded minimum-plus-five event window. (b) Mean CPU seconds per reaction for matching, including reduction, and decoding. Both directions and the full cut sweep contribute; publication and external verification are excluded.

search directions and the cut sweep raises its coverage to 96.92%. These results show the value of broader correspondence discovery before selecting one answer.

Table 4 Golden reference coverage across AAM methods. All methods are evaluated over 1,851 reactions. Coverage counts reference inclusion among returned alternatives; a single prediction is a singleton set. MAPPA uses the decoded minimum-plus-five output of the full one- and ten-seed studies. Reference checks follow Section 3.2. Expanded SLAP includes both weight modes, both directions and the cut sweep.

Method	Search setting	Output	Coverage
MAPPA	**1 seed, sweep**	**Families**	**1,833 (99.03%)**
MAPPA	**10 seeds, sweep**	**Families**	**1,842 (99.51%)**
SLAP	Expanded sweep	Candidates	1,794 (96.92%)
SLAP	Binary, both directions	Candidates	1,620 (87.52%)
SLAP	Binary, default	Candidates	1,591 (85.95%)
LocalMapper	Default	Single mapping	1,605 (86.71%)
Chython	Default	Single mapping	1,561 (84.33%)
RXNMapper	Default	Single mapping	1,547 (83.58%)
RDT	Default	Single mapping	1,030 (55.65%)
Indigo	Default	Single mapping	692 (37.39%)

4.2 Computational cost and seed scaling

One-seed MAPPA requires an average of 3.19 CPU seconds per Golden reaction: 1.31 for matching and 1.88 for decoding. EnzymeMap requires 13.16 CPU seconds, divided into 4.41 for matching and 8.75 for decoding (Table 5). The EnzymeMap median is 5.80 seconds and its 95th percentile is 41.86 seconds, indicating that a minority of difficult reactions account for much of the total computation.

Table 5 Mean CPU seconds per reaction. Matching includes fragment discovery and reduction. Both directions and the full cut sweep are included. Decoding is measured separately; publication and external verification are excluded.

Study	Matching	Decoding	Total
Golden, 1 seed	1.31	1.88	3.19
Golden, 10 seeds	11.64	9.28	20.92
EnzymeMap, 1 seed	4.41	8.75	13.16

Increasing Golden from one to ten seeds raises matching cost by 8.87-fold and decoding cost by 4.95-fold. Matching scales approximately linearly with seed count over this range. Decoding grows less because repeated fragments and branch compositions share their event responses. The additional exploration recovers nine more references at an average total cost of 20.92 CPU seconds per reaction.

The paired timing study compares computational cost on the same inputs and CPU model (Figure 6). Standard binary SLAP is fastest. MAPPA requires less computation than expanded SLAP, including its separate decoding stage. Detailed measurement scopes are given in Appendix B.

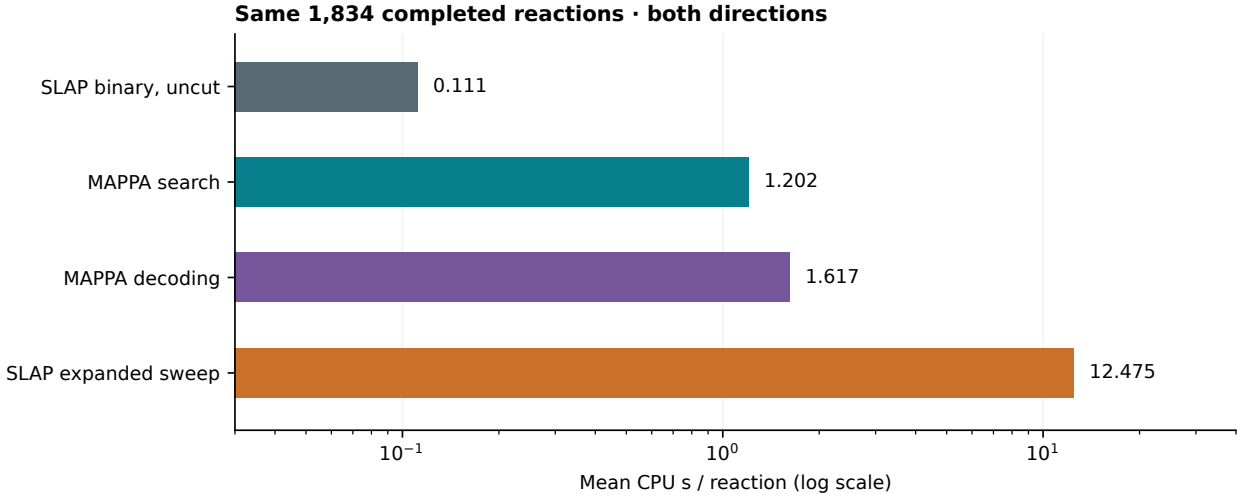


Figure 6 Computational cost under paired evaluation. Mean CPU time on the 1,834 Golden reactions completed by all protocols, measured on the same CPU model. Standard SLAP is binary and uncut; expanded SLAP combines binary and weighted modes with the cut sweep. All protocols use both directions. MAPPA matching and decoding are shown separately; the time axis is logarithmic.

4.3 Reference event counts relative to the detected minimum

The saved GT-consistent mappings have the same event count as the detected minimum for 1,630 Golden reactions (88.06%) and 44,086 EnzymeMap reactions (91.90%; Figure 7). Most other mappings lie within five events above the minimum. These distributions show that low event counts often agree with the reference, while retaining nearby alternatives provides additional reference-consistent choices.

Each offset describes a concrete reference-consistent mapping, including its saved hydrogen completion. It does not minimize over alternative completions of the reference.

Anchored completion succeeds for all 196 previously absent references, included in Figure 7. The completed count equals the original detected minimum for nine Golden and 83 EnzymeMap cases; nine EnzymeMap completions lie two events below it. The remaining completions lie above the detected minimum, reaching offsets of 12 in Golden and six in EnzymeMap. Every supplied pair is preserved, and each selected completion's model count agrees with direct counting from the original bond matrices.

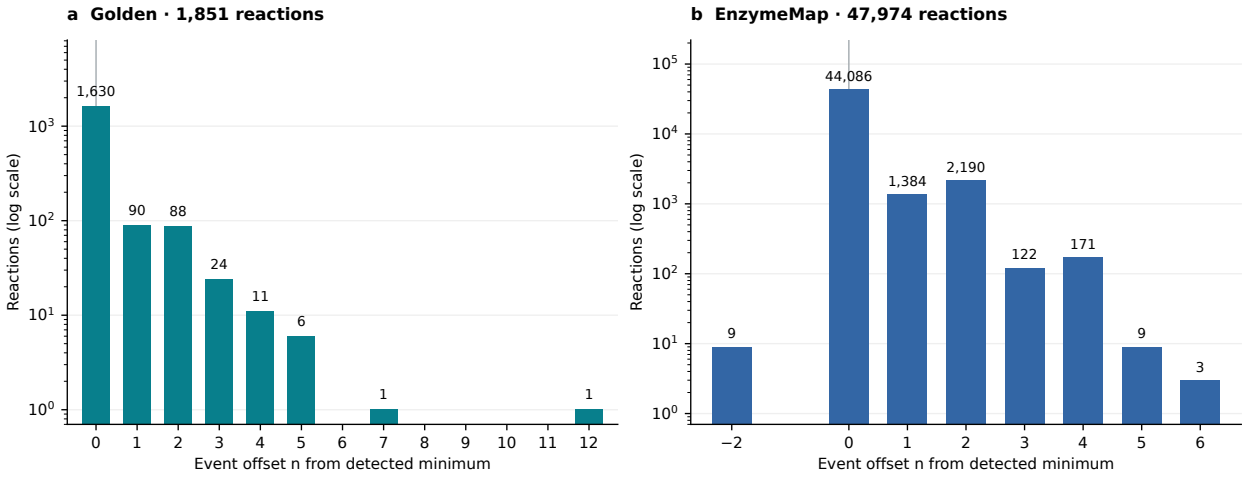


Figure 7 Reference-mapping event counts relative to the detected minimum. Each reaction contributes $n = E(m_{\mathrm{GT}}) - U$, using a verified GT-consistent full mapping and the original saved detected minimum. The lowest recorded verified count across both directions is used, with no additional optimization. Saved anchored completions supply the 18 Golden and 178 EnzymeMap references absent from the original output. Both terms use the same all-atom decoder model. Labels give reactions per bin; the vertical axis is logarithmic. The nine negative EnzymeMap offsets are anchored completions two events below the original detected minimum. Independent coverage is unchanged.

4.4 How many reaction cores remain at the minimum?

An event count describes how many bonds change, but does not identify those bonds. We define a *reaction core* as the pattern of bonds that break, form, strengthen or weaken under a mapping, including changes involving hydrogen. For example, two mappings may each break one bond and form one bond, yet act on different atom pairs. They then describe different net transformations despite having the same event count. Mappings with the same bond-change pattern count as one core, after accounting for equivalent atom labels, even if they arise from different fragment compositions.

We count these distinct patterns among the saved mappings at each reaction's detected minimum event count. In the one-seed Golden results, 1,252 of 1,851 reactions have one observed core (67.64%; Figure 8). The median is one core, the mean is 2.97, and the 95th percentile is four; the largest count is 522. Thus, most reactions retain few observed bond-change alternatives at the minimum, while a small minority remain highly ambiguous.

Minimum-event selection can therefore leave several net transformations for subsequent chemical selection. A reaction core specifies endpoint bond changes; it leaves the sequence of elementary steps unspecified. The counts describe observed alternatives in the saved mappings, without further enumeration of internal symmetry choices.

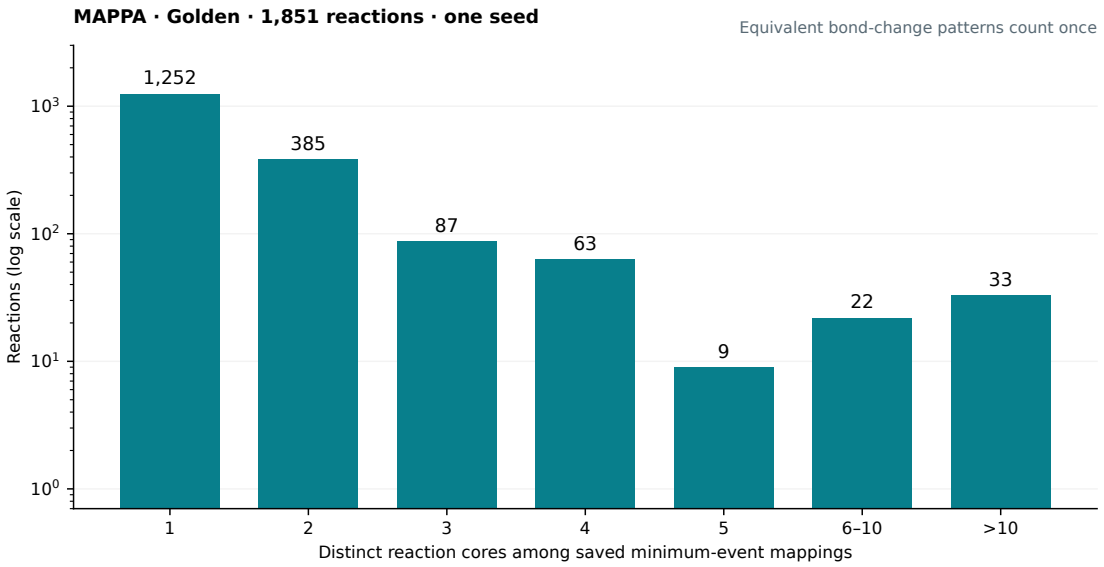


Figure 8 Distinct reaction cores among saved minimum-event mappings. Counts cover all 1,851 one-seed Golden reactions and combine both search directions. Each core identifies bonds whose orders decrease or increase, including hydrogen events. Symmetry-equivalent atom labels count once, and different fragment partitions with the same bond-change pattern are merged. The vertical axis is logarithmic.

5 Discussion

MAPPA combines high reference coverage with explicit retention of mapping alternatives. Its fragment representation organizes these alternatives around shared structural constraints, allowing the same local relationship to contribute to many branches. Matching discovers those constraints, and decoding connects their permitted rearrangements to bond changes. This separation supports both the generation of candidate correspondences and the interpretation of how those candidates differ.

The seed study illustrates a practical trade-off between exploration and computation. One seed recovers most references, while additional seeds provide a modest coverage gain with approximately proportional matching cost. The cut sweep offers a complementary way to vary the constraints encountered during greedy growth. Together, these controls allow search effort to be adjusted to the application, with independent searches available for parallel execution.

The event distributions support using low-event mappings as a shortlist, but minimum bond-event count is not a reliable stand-alone surrogate for the chemically correct mapping of an overall reaction. Overall reactions may comprise multiple elementary steps: the net bond changes between their endpoints do not describe the intervening mechanism, and the correct atom correspondence need not minimize those changes. Selecting only the smallest count can therefore discard a chemically meaningful higher-event mapping. MAPPA retains nearby alternatives so that reaction-specific chemical information can guide the final choice.

Retained alternatives could also assist reaction-path preparation. Atom correspondence determines how endpoint coordinates are paired during interpolation (3); different assignments can therefore supply distinct initial paths for methods such as the nudged elastic band (6). Coupling these alternatives to chemical ranking and path refinement could help select correspondences using information beyond endpoint structure. Beyond pathway preparation, MAPPA could support reaction-template extraction from mapped bond changes. It could also serve as a structural verifier during reinforcement-learning training of large language models, providing feedback on proposed atom correspondences and bond changes.

Code and data

Code and reproduction instructions are available at MAPPA source repository, branch `main`. The repository guide `manuscript/REPRODUCIBILITY.md` links full benchmark configurations, software versions, per-case results, and figure-generation scripts. Golden and EnzymeMap are distributed by their original authors (8; 5).

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A Technical algorithm details

The complete stage sequence is summarized in Algorithm A1. The following sections specify the constraints, symmetry operations and event model used by this workflow.

Algorithm A1 Matching (including reduction), followed by decoding

Require: Endpoint graphs, discovery policies, seed/cut plan, branch cap and event window
for all requested search directions **do**
  for all cut conditions and seed orderings **do**
    Initialize the conditional search hierarchy
    while a retained continuation has an eligible fragment seed **do**
      Grow using one-witness existence queries
      At saturation, construct distinct compressed placements
      Continue each placement with earlier assignments locked
      Prune and recount the hierarchy from leaves to root
    end while
    Retain terminal compositions and compact provenance
  end for
  Share fragments by atom-set pair; reduce repeated branch compositions
end for
Complete matching, including reduction, before batch decoding
for all reaction/direction catalogues **do**
  Inventory unique fragments and prepare their shared symmetry-event responses
  Compose branch contributions and the sweep-endpoint correction
  Retain queryable branches within the requested event window
end for

A.1 Growth and completion constraints

Let V_B, V_T be the base and target atom sets, Z_B, Z_T their elements, and W_B, W_T their original bond-order matrices. Growth seeks an injective partial correspondence $m : D \subseteq V_B \to V_T$ that respects locked assignments. Every mapped atom preserves its element, and every accepted growth edge $(u, v) \in A$ satisfies

$$Z_B(u) = Z_T(m(u)), \qquad W_T(m(u), m(v)) \geq b, \qquad |W_B(u, v) - W_T(m(u), m(v))| \leq \tau_{\text{iso}}. \tag{5}$$

Only accepted growth edges belong to A. Nonedges and other bonds between the included atoms do not add matching constraints: this is non-induced relaxed subgraph isomorphism.

An existence query excludes occupied targets, removes destinations without compatible neighbors, and checks whether the remaining domains admit distinct targets. If these constraints do not decide the query, it searches the remaining assignments and stops at one witness. The same constraints establish absence when all choices are excluded. A witness may rearrange the growing fragment while earlier fragments remain locked. Compatible saved fragment witnesses can answer a query directly, provided they satisfy the current occupancy, anchors and accepted edges; reuse does not replace the new trajectory's random growth decisions.

At saturation, completion seeks distinct target atom sets, excluding each represented set from subsequent search. Internal assignments on one atom set remain compressed. A certified global target symmetry may additionally group placements when it preserves the entire remaining matching problem: the original weights, active connectivity, locks and discovery predicates. Otherwise the placements remain separate. Local fragment symmetry alone cannot establish that two global placements are interchangeable.

For each fragment side, nauty computes automorphism generators of its element-colored binary graph, with anchors fixed individually. Whole-ring or whole-arm moves remain correlated operations; terminal exchanges may be independent. Binary symmetry does not enforce bond-order tolerance. Discovery tolerance applies to the starting correspondence; subsequent local automorphisms on both sides are evaluated for their bond events using the original weights. Ordered global placement operations remain in their saved order.

A.2 Hierarchical branching and fragment identities

Each completed placement creates a child continuation. After a completed step, count surviving continuations from leaves to root. At an over-limit node, discard its largest child subtree, recount, and repeat until the retained count satisfies the cap; ties favor the earlier decision. The parent is then assessed using the reduced count. A separate completed-fragment limit rejects placement expansions exceeding its allowance. Internal symmetry permutations do not count as additional branches. Both limits restrict search coverage.

The final matching subphase unifies fragments by their base/target atom-set pairs and assigns shared identifiers. One original fragment record is retained intact for each pair. Branches are deduplicated by their fragment composition and ordered placement context, while alternative cut and partition contexts are retained for decoding. This is structural deduplication, without symmetry optimization or event queries. Equal atom sets define a shared identity; they do not authorize arbitrary atom correspondences.

A.3 Local event responses and branch composition

Each unique fragment is prepared once. Its response retains a selected starting correspondence, compact binary automorphism generators and the original-weight contributions to bond events. Operations are separated only when no variable event or required boundary endpoint couples them. An arm exchange therefore preserves the joint motion of that arm's atoms. Event-invariant operations share computation, while variable contributions retain their dependence on the relevant choices; the response does not store a Cartesian table of complete mappings.

Two relaxations provide lower bounds for local optimization. One compares bond-weight multisets within element-pair types without imposing atom consistency. The other matches incident bond assignments and then atoms injectively; because each internal event touches two atoms, half the incident cost, rounded up, is a lower bound. The stronger bound is used. These bounds need not be attainable.

First improve the saved correspondence under its fragment automorphisms. Search for another starting correspondence only when growth and final internal connectivity differ on either side and the current event count remains above an established bound. New starting correspondences must satisfy the saved discovery constraints. Subsequent binary symmetry choices are evaluated for events without reimposing discovery tolerance. Preparation and improvement share the per-fragment budget; attaining the bound ends optimization. Otherwise the response retains the best found witness and its bound. A minimum within the selected symmetry orbit is not a guarantee over every possible starting correspondence.

Branch assembly adds the disjoint fragment contributions and the fixed ordinary-boundary contribution. The single-edge sweep requires a joint correction for its endpoint choices, which can lie in neighboring fragments. That correction reuses the prepared local responses, preserves their dependence on internal events and contributes zero to two events. When internal costs are constant, only jointly reachable endpoint pairs need examination.

The decoder tracks each branch's lower bound and an attained count U. If compatible local minima attain the smallest pre-sweep composition L, the sweep correction gives $U \leq L + 2$. Only compositions capable of improving U need refinement; afterward, branches with lower bounds above $U + n$ are excluded. Unfinished local optimization retains its lower bound and an actual witness count rather than assuming the bound is attainable. Saved responses support subsequent queries within the window; a count interval does not imply that every integer inside it is attainable.

Ordinary boundaries use the independent-fragment approximation described in the main text. Consequently, the model event count can differ from a direct whole-reaction count. Reference-window membership uses the model count; physical bond changes are evaluated separately when constructing reaction cores.

A.4 Molecular examples

The introductory EnzymeMap example (case 13175, Figure 2) fixes two different heavy-atom correspondences: four heavy-atom events for the reference and two for the alternative. The endpoints contain nine heavy atoms and six hydrogens. Checking all $6! = 720$ hydrogen assignments for each fixed heavy-atom correspondence

gives six total events in both cases. For the reference, transfer of a carboxyl OH hydrogen contributes two additional events; for the alternative, transfer of both formaldehyde hydrogens contributes four. The example therefore shows distinct atom origins that remain tied after hydrogen completion, not a lower all-atom count for the alternative.

The constructed matching diagrams illustrate graph relationships. In Figure 3, the two target rings have different attachments and therefore define two distinct placements; each admits twelve internal rotations and reflections. Hydrogen constraints are omitted from this illustration. In Figure 1, exchanging the acetate oxygen images adds two internal events to two boundary events at the 0.5 threshold. These examples use formal bond orders. Their endpoint graphs and direct bond-change checks are supplied with the figure sources.

B Evaluation details

B.1 Benchmark settings and reference checks

The settings below apply to the current Golden and EnzymeMap studies. Golden uses one or ten seed orderings per cut; EnzymeMap uses one. Every search includes both directions, the uncut condition and all eligible single-edge cuts.

Table A1 Shared benchmark settings. Events use the original bond orders; binary fragment graphs determine the allowed symmetry operations.

Parameter	Value
Bond representation	Prepared formal bond orders, explicit hydrogens
Discovery tolerance τ_{iso}	2
Graph/cut presence threshold b	0.2
Hierarchical branch cap	100
Random seed	42
Bond-event threshold	0.5 organic; 0.3 metal-related
Local preparation/improvement budget	5 seconds per unique fragment
Decoded event window	$[U, U+5]$, with U the best attained model count

An event requires a change strictly greater than its threshold. The empirical 0.3 threshold applies when at least one atom is a metal; 0.5 is the non-metal default. Reference checking preserves annotated heavy-atom pairs and unmatched-atom requirements. Unannotated hydrogen assignments must admit a complete injective witness, replayed through the saved response and independently counted inside the model window. The annotated reference does not guide the benchmark search.

Comparator outputs are evaluated over the same 1,851 Golden reactions, including invalid or failed outputs in the denominator. Single-answer methods contribute one candidate; SLAP contributes its returned candidates. Comparator checks permit chemically equivalent endpoint labels while preserving charge, isotope, hydrogen count and stereochemistry. Expanded SLAP combines binary and weighted modes, both directions and the single-edge sweep. Software versions, adapters, per-case outcomes and input identities are recorded in the repository's reproducibility guide.

B.2 Timing scope

Full-study CPU totals sum successful stage computation over both directions and every cut. Matching includes reduction; decoding is separate. These totals include stage loading and result construction, but exclude publication and external verification. Golden combines measured matching and postprocessing passes; EnzymeMap combines completed original tasks and completed recoveries. The totals measure accumulated

computation, not uninterrupted elapsed time. Seed scaling compares the measured workloads rather than isolating a cache effect.

The paired timing comparison uses an Intel Xeon Gold 6442Y CPU and SLAPMapper revision `ea248fd`. All three protocols use identical prepared inputs, and their reported means use the same 1,834 completed reactions. These paired timings exclude loading, publication and external verification. Coverage retains all 1,851 reactions as its denominator. Hardware-controlled speed comparisons use this paired study, not ratios between the full Golden and EnzymeMap campaign means.

B.3 Event offsets and reaction-core counting

Each of the 49,825 reactions contributes one offset: the lowest saved verified reference-consistent model count across directions minus the original detected minimum. Unannotated hydrogen assignments are those of the selected witness. The 196 missing references use separately obtained anchored completions; those do not increase independent coverage. No branch-minimum or additional symmetry search is performed for the plot. The selected completion need not minimize over every possible hydrogen assignment.

Reaction-core counting uses saved witnesses at the detected minimum for all 1,851 one-seed Golden reactions. Both directions are converted to the same atom indexing. Direct comparison of the original matrices identifies signed bond changes, including hydrogen events; equivalent atom labels and different fragment partitions with the same pattern count once. This counts observed reaction cores in saved witnesses, without enumerating additional symmetry alternatives. A core describes net bond changes, not a complete sequence of elementary steps.

Figure A1 gives the percentage form of the event-offset distribution in Figure 7.

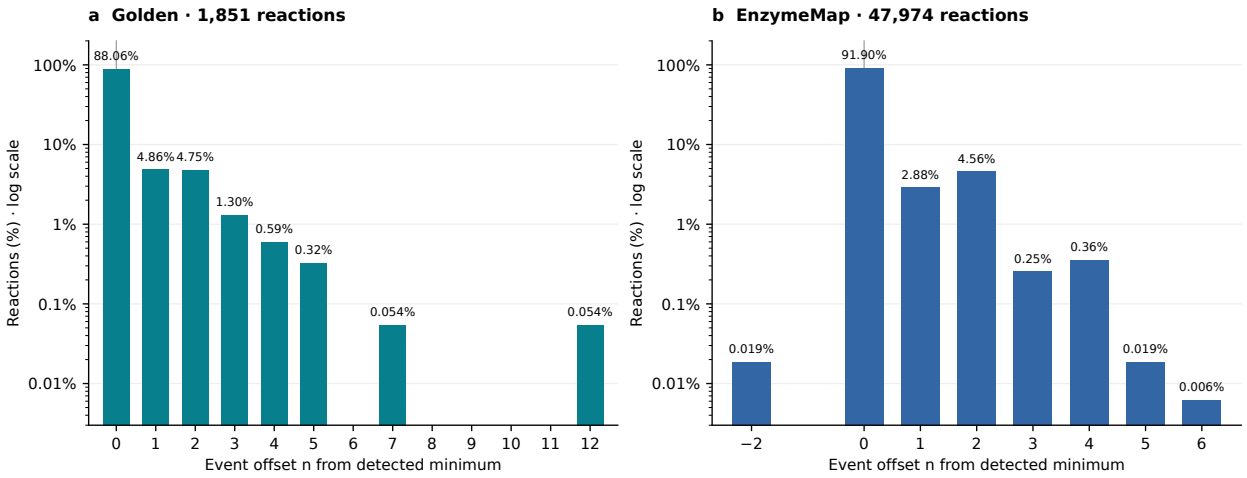


Figure A1 Reference event-offset distribution as percentages. The same verified mapping counts and detected minima as in Figure 7, expressed as a percentage of all 1,851 Golden or 47,974 EnzymeMap reactions. Saved anchored completions are included. Each dataset sums to 100% before rounding. Bar labels give percentages; the vertical axis is logarithmic.

The repository's `manuscript/REPRODUCIBILITY.md` links the source versions, benchmark evidence, mapping certificates and figure-generation inputs supporting these analyses.